A Simplified Round-by-round Soundness Proof of FRI

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ZKS

Introduction

Fast Reed–Solomon IOPP (FRI)
FRI [BBHR18] is an interactive oracle proof of proximity (IOPP) for Reed–Solomon codes: the prover convinces the verifier that a function $f_0 : \mathcal{L}_0 \to \mathbb{F}$ is close to the evaluations of a low-degree polynomial.

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What is FRI? **Can** we come up with a simpler proof? **Why** should we care?

Agenda

Motivation: SNARKs

Background: RS codes and IOPPs

The FRI protocol

Analysis: soundness of FRI

SNARKs

SNARKs

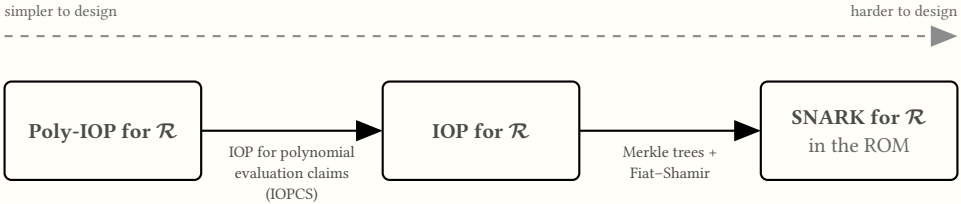
- SNARK: succinct, non-interactive argument of knowledge.
- A short string that certifies that the prover knows a witness to some NP statement.
  - using elliptic curves, $\approx$ 260 B.
  - using lattices, $\approx$ 60 KB.
  - using only hash-functions, $\approx$ 100-1000 KB.
- Widely deployed in the blockchain industry, growing adoption from tech giants.

A recipe for hash-based SNARKs

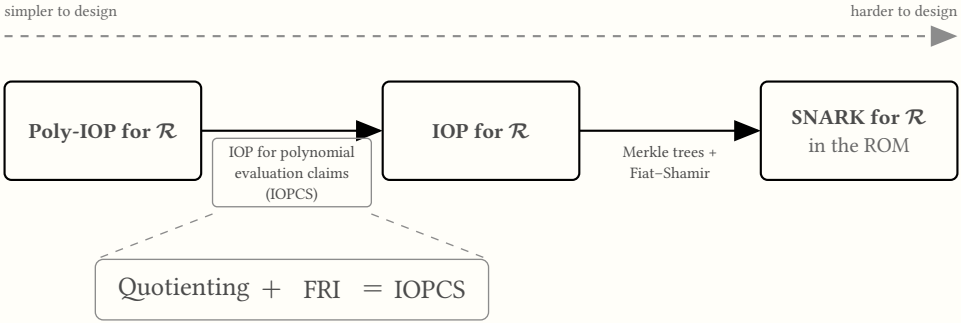
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SNARK for $\mathcal{R}$
in the ROM

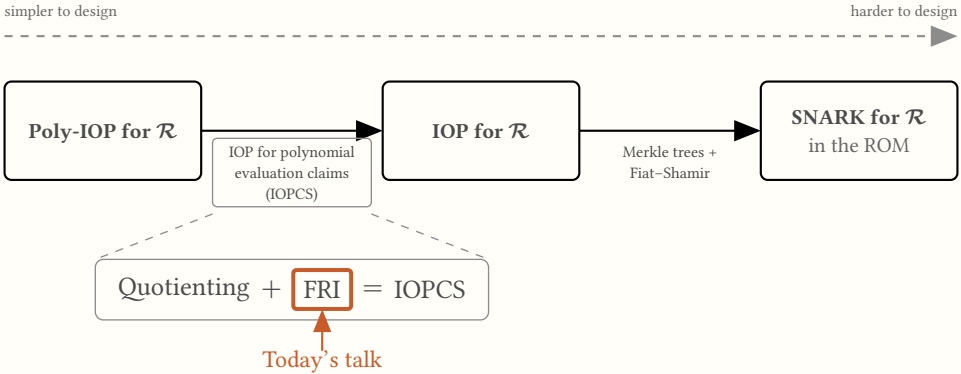
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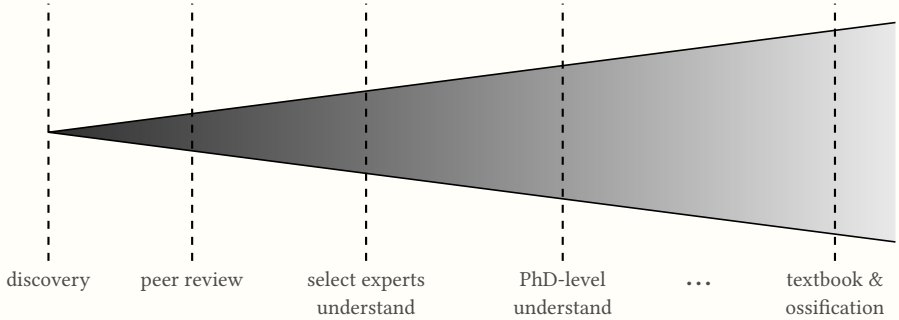


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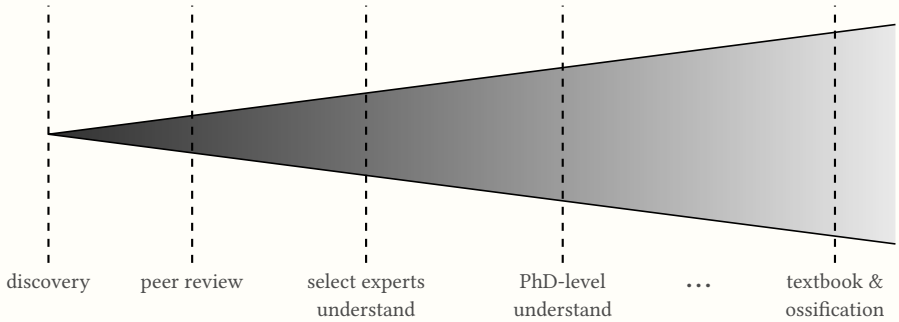


On simplifying proofs

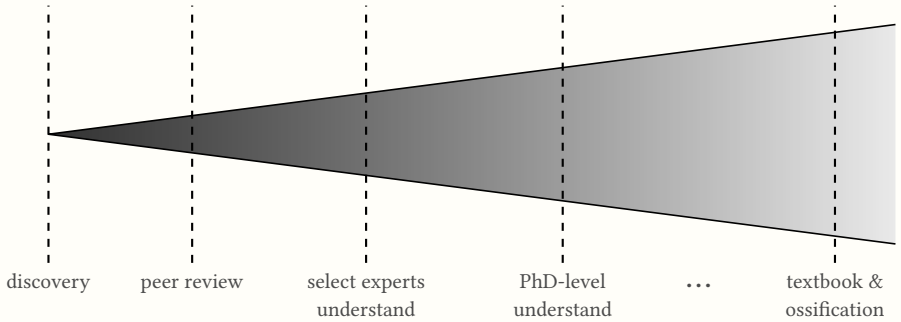
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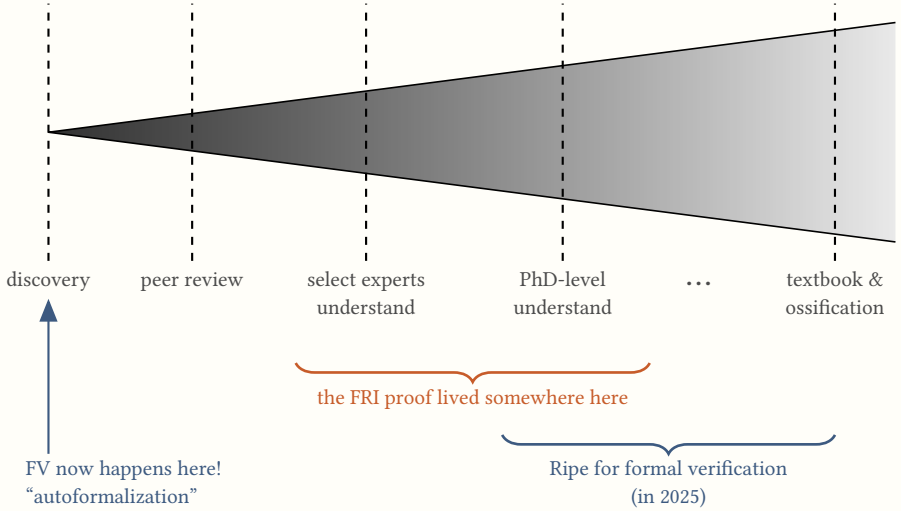
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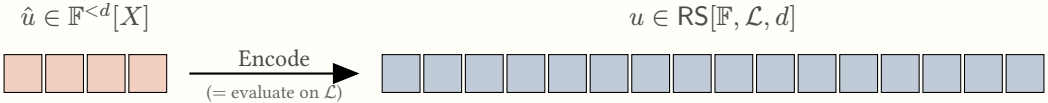
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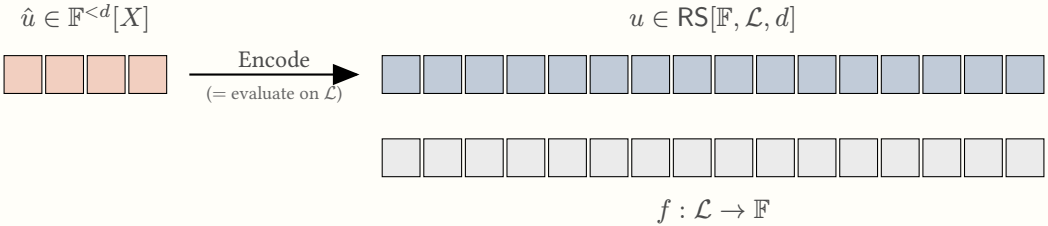
Reed–Solomon codes

A message of length d is a polynomial $\hat{u} \in \mathbb{F}^{<d}[X]$; its codeword $u \in \mathsf{RS}[\mathbb{F}, \mathcal{L}, d]$ is the evaluations of $\hat{u}$ on a domain $\mathcal{L} \subseteq \mathbb{F}$.



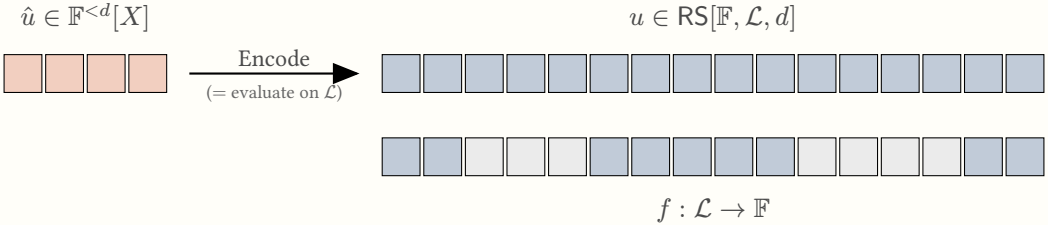
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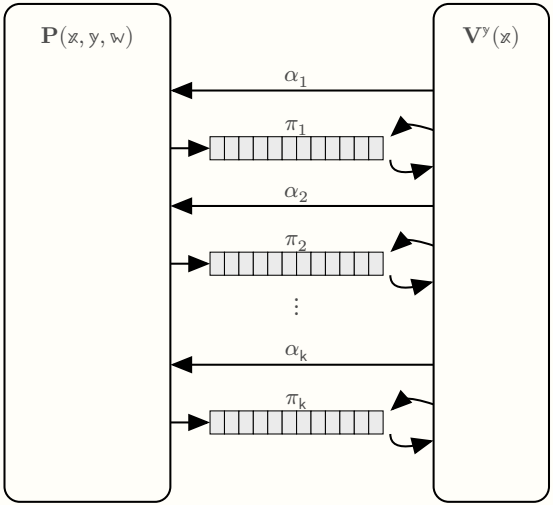
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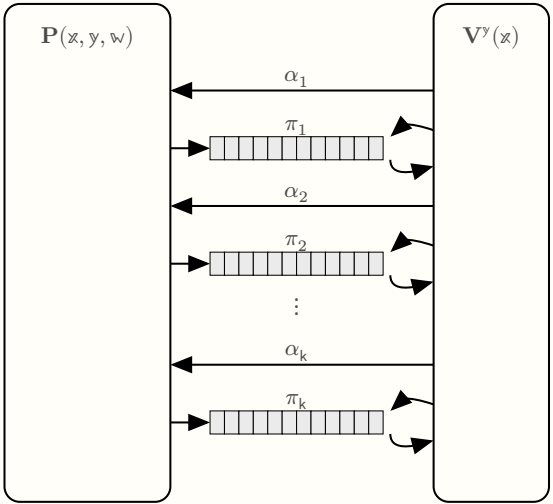


- Write $\Delta(f, g)$ for the relative Hamming distance.
- The agreement set between f and u is the subset $S \subseteq \mathcal{L}$ where f and u agree.

Interactive oracle proofs (IOPs)

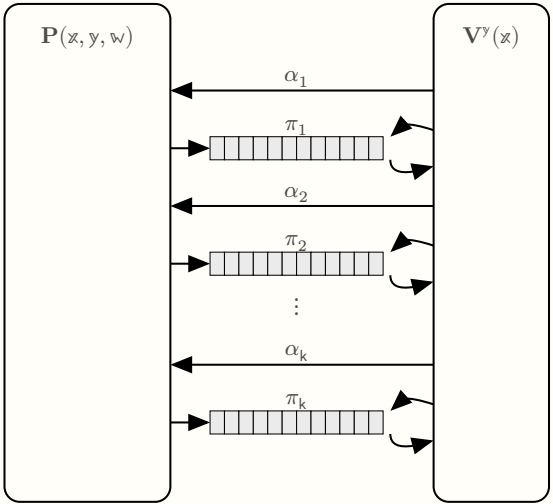


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- Prover messages are oracles: the verifier reads only the positions it queries.
- Statement split into $\mathbb{x}$ (read in full) and $\mathbb{y}$ (oracle access). The prover knows a witness $\mathbb{w}$.
- Public coin: verifier messages are uniformly random and independent, including the query positions.
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  $\Longrightarrow$ SNARK size depends on the number of queries.

Figure adapted from Chiesa and Yogev, *Building Cryptographic Proofs from Hash Functions*, Fig. 23.1 (CC BY-SA 4.0).

IOP of proximity (IOPP)

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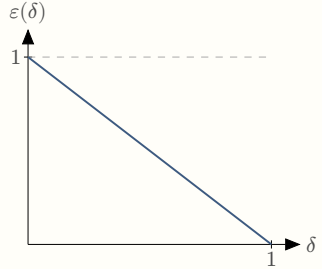
Definition (IOPP for $\mathcal{C}$, informal)
An IOP for the relation $\mathcal{R}_{\mathcal{C}}$ where soundness is a function of distance.

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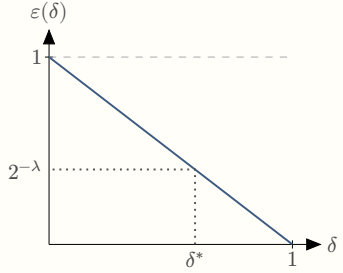
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- We say that the IOPP has λ bits of security at distance δ^* if

$$\forall f \text{ s.t. } \Delta(f, \mathcal{C}) \geq \delta^*, \quad \Pr\left[\mathbf{V}^f = 1\right] \leq 2^{-\lambda}$$



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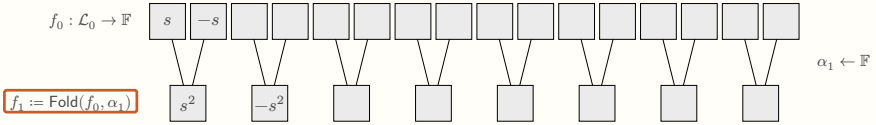
FRI in pictures: folding phase

$f_0 : \mathcal{L}_0 \to \mathbb{F}$ 

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The Fold function
For every $\alpha \in \mathbb{F}$, $\mathsf{Fold}(\cdot, \alpha)$:

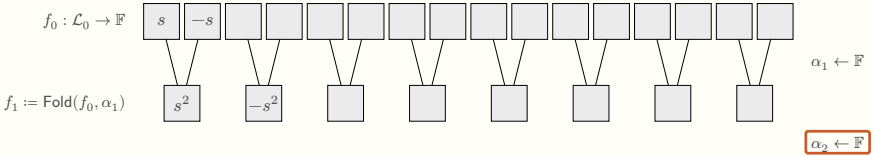
- **maps $\mathcal{C}_0 := \mathbf{RS}[\mathbb{F}, \mathcal{L}_0, d_0]$ to $\mathcal{C}_1 := \mathbf{RS}[\mathbb{F}, \mathcal{L}_1, d_1]$**, where

$$\mathcal{L}_1 := \mathcal{L}_0^2 = \{x^2 : x \in \mathcal{L}_0\} \quad \text{and} \quad d_1 := d_0/2.$$

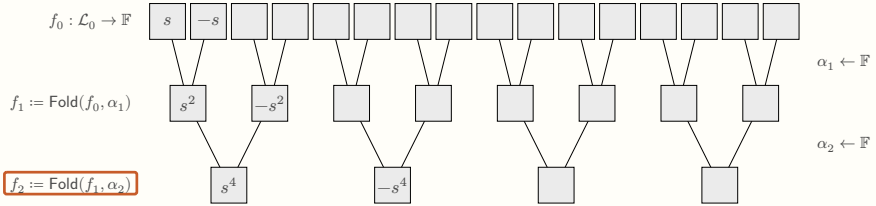
  Domain size and degree bound are both halved. The rate $\frac{d_1}{|\mathcal{L}_1|} = \frac{d_0}{|\mathcal{L}_0|} = \rho$ is unchanged.

- **is locally computable**. *E.g.*, $\mathsf{Fold}(f_0, \alpha_1)(s^2)$ is computed from $f_0(s)$ and $f_0(-s)$.

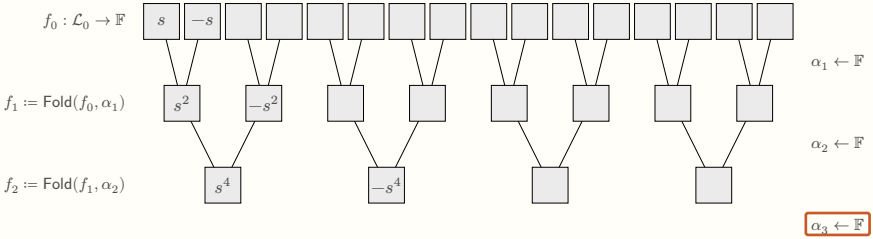
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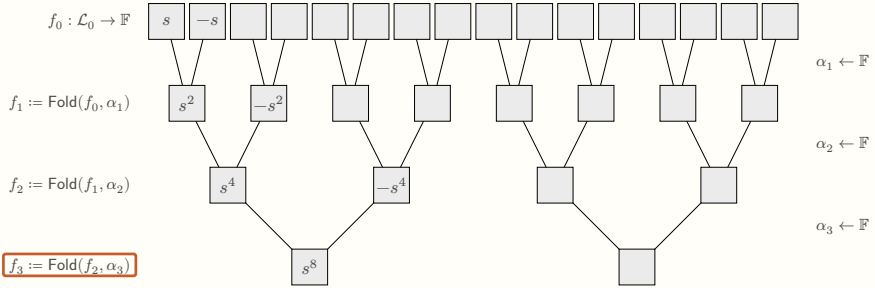
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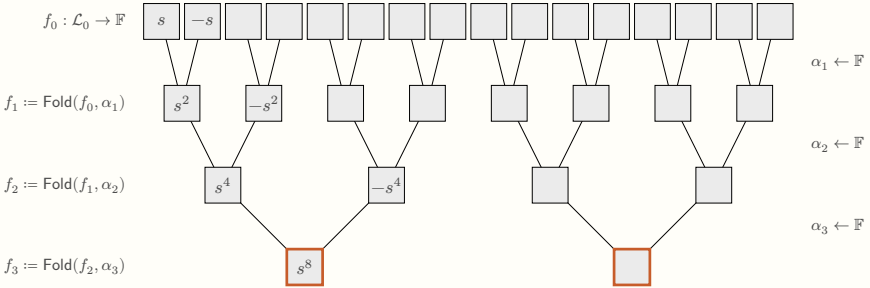
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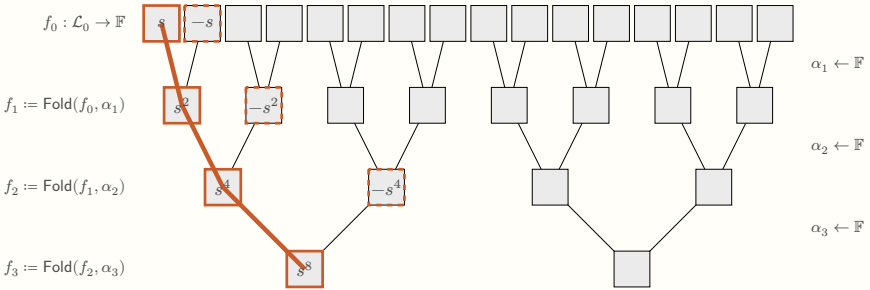


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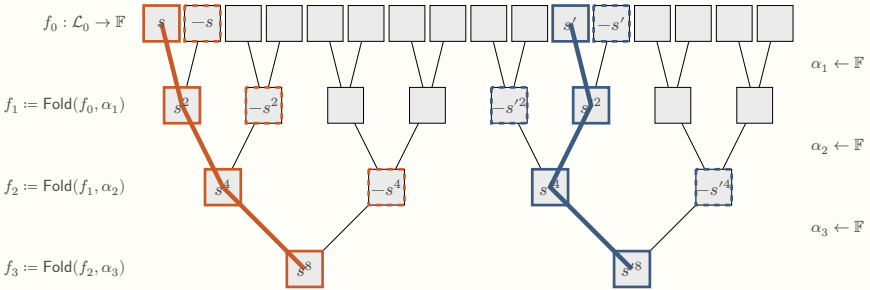
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3. **Repeat.** A second query s' induces a second path.

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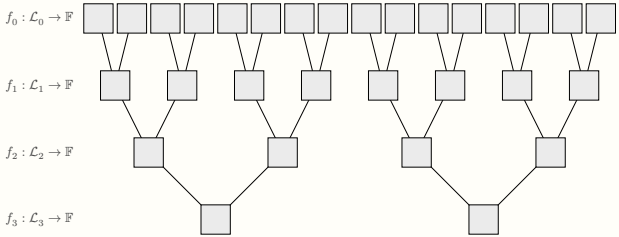
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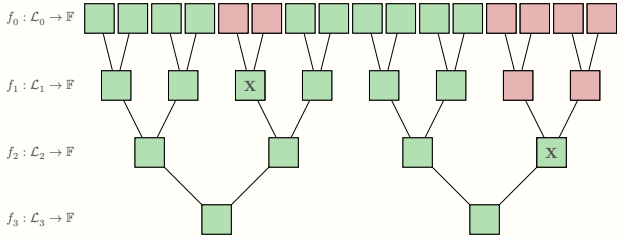
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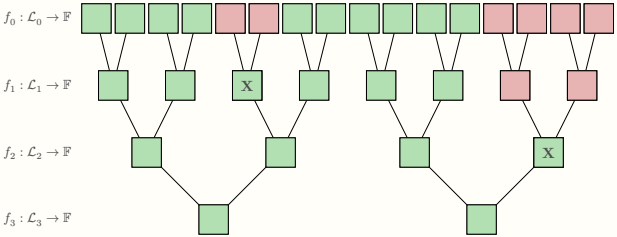
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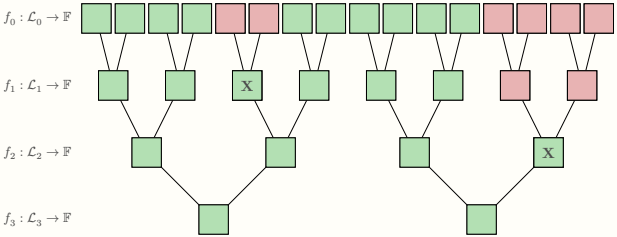


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3. Relate the # of green nodes in $\mathcal{L}_0$ to $\Delta(f_0, \mathcal{C}_0)$, leveraging properties of Fold.



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If there is a large enough set of green nodes where f_i agrees with $\mathcal{C}_i$ then, f_{i-1} agrees with $\mathcal{C}_{i-1}$ on green nodes except with negligible probability over the choice of α_i.

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Case 1: at least $(1-\theta)$ fraction of green nodes.

$\Rightarrow$ since P_θ holds, agreement propagates upwards.

$\Rightarrow$ exists u_0 that agrees with f_0 on green nodes.

$\Rightarrow$ accept probability is bounded by $(1-\delta)^t$.

Case 2: less than $(1-\theta)$ fraction of green nodes.

Proving 3., continued

$$\begin{aligned}\Pr[\mathbf{V} \text{ accepts}] &\leq \Pr[P_\theta \text{ fails at some round}] + \Pr[\mathbf{V} \text{ accepts} \mid P_\theta \text{ holds at all rounds}] \\ &\leq \underbrace{\sum_i \Pr[P_\theta \text{ fails at round } i]}_{\sum_i \varepsilon_i^{\mathsf{fold}}(\delta)} + \underbrace{\Pr[\mathbf{V} \text{ accepts} \mid P_\theta \text{ holds at all rounds}]}_{\varepsilon^{\mathsf{qry}}(\delta)}\end{aligned}$$

$\varepsilon_i^{\mathsf{fold}}$ is a probability over α_i. Property of $\mathcal{C}_{i-1}, \mathcal{C}_i$ and Fold.

↓ Lemma 5.1

Property of linear combinations of $\mathcal{C}_i$: mutual correlated agreement [ACFY24].

↓

Write $\varepsilon^{\mathsf{mca}}$, give the problem to coding theorists. See the Proximity Prize and [ABF26].

$\varepsilon^{\mathsf{qry}}$: case analysis.

Case 1: at least $(1-\theta)$ fraction of green nodes.

$\Rightarrow$ since P_θ holds, agreement propagates upwards.

$\Rightarrow$ exists u_0 that agrees with f_0 on green nodes.

$\Rightarrow$ accept probability is bounded by $(1-\delta)^t$.

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Join the cases by taking the max.

The theorem, as it appears in the paper

Theorem (Theorem 5.2)
Consider FRI with parameters as in Construction 4.4 and assume Powers is a proximity generator with mutual correlated agreement (cf. Definition 2.2) with error $\varepsilon^{\mathsf{mca}}$. For every tradeoff parameter $\theta \in [0,1]$, FRI has round-by-round soundness with error functions $\varepsilon_1^{\mathsf{fold}}, \ldots, \varepsilon_k^{\mathsf{fold}}, \varepsilon^{\mathsf{qry}} : (0,1] \to [0,1]$ such that for every initial function $f_0 \notin \mathcal{C}_0$:

- for all $i \in [k]$, $\varepsilon_i^{\mathsf{fold}}(\delta) = \varepsilon^{\mathsf{mca}}(\mathcal{C}_i, m_i, \theta)$,
- $\varepsilon^{\mathsf{qry}}(\delta) = (\max\{1-\theta, 1-\delta\})^t = (1 - \min\{\theta, \delta\})^t$,

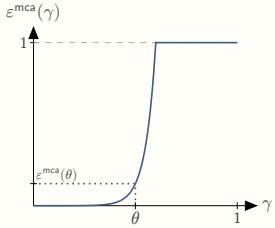
where $\delta := \Delta(f_0, \mathcal{C}_0)$.

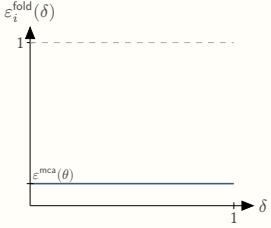
Understanding the error bounds

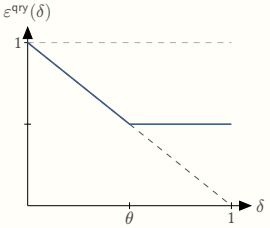
Understanding the error bounds

The tradeoff parameter

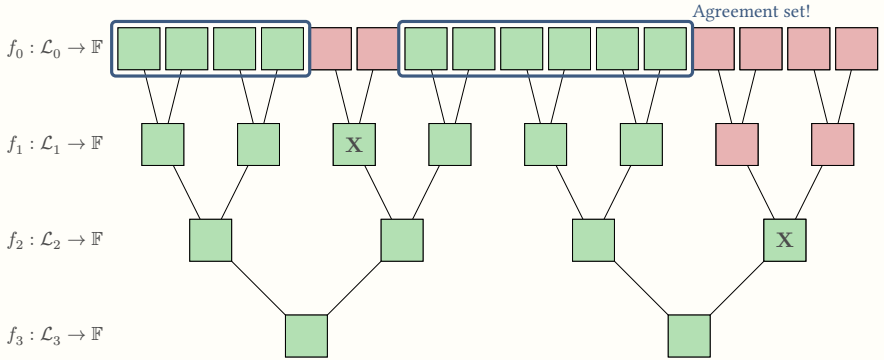
- Folding round errors: $\varepsilon_i^{\mathsf{fold}}(\delta) = \varepsilon^{\mathsf{mca}}(\mathcal{C}_i, 2, \theta)$ grows with θ — wants θ small.
- Query round error: $\varepsilon^{\mathsf{qry}}(\delta)$ is clamped from below at $(1-\theta)^t$ — wants θ large.







Bonus: extraction from erasure decoding



- If $\Pr[\mathbf{V} \text{ accepts}] \geq (1-\theta)^t$ and $\theta \leq 1-\rho$, case 1 gives a *unique* $u_0 \in \mathcal{C}_0$ agreeing with f_0 on the green top nodes: $\geq \rho\ |\mathcal{L}_0| = d_0$ points.
- Recover u_0 by erasure decoding (interpolation on green nodes), no list decoding. The folding phase is a binding commitment, beyond the unique decoding radius.

Comparison to previous proofs

Our bounds are the same as previous proofs [BCIKS20, ABN23]; ethSTARK [StarkWare21] observed that these are round-by-round bounds. The difference is the guarantee given by a *good* folding challenge:

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If all green layers are large enough, agreement propagates upwards through the green nodes.

CA (previous proofs)
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If all green layers are large enough, agreement propagates upwards through the green nodes.

CA (previous proofs)
If all green layers are large enough, agreement propagates upwards *at unknown nodes*, possibly not aligned with green.

- Losing the location forces previous proofs to introduce *weighted* distances and a weighted correlated agreement theorem, with weights tracking the quality of agreement at each node.
- Without keeping track of agreement sets, extraction requires error decoding.

Summary

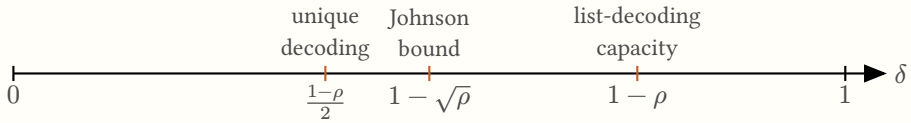
- FRI is part of the SNARK pipeline.
- We present a simpler proof that provides new insights.
- Proof overview:
  - define and bound a bad event for honest folds.
  - if many dishonest folds, verifier rejects.
  - else, if many honest folds and no bad event, acceptance probability is related to distance δ.
- Paper also includes: formal proof, formal statement of the new insights, comparison to previous works and batched FRI.

Thank you! Questions?

Full paper: ia.cr/2025/1993.

Backup: decoding regimes

How far can f be from $\mathcal{C} = \mathsf{RS}[\mathbb{F}, \mathcal{L}, d]$ and still have few codewords nearby? ($\rho = 1/4$)

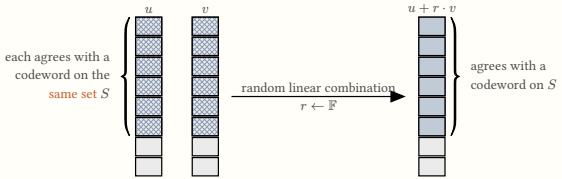


State-of-the-art before the Proximity Prize:

- Below $\frac{1-\rho}{2}$: at most one codeword within distance δ of f.
- Below $1-\sqrt{\rho}$: polynomially many codewords (Johnson bound).
- Between $1-\sqrt{\rho}$ and $1-\rho$: unknown area.
- Above $1-\rho$: vacuous statement.

Our analysis has a parameter θ that plays the role of "how far we dare to go".

Backup: mutual correlated agreement (MCA)



1. **Commit.** Fix two functions $u, v : \mathcal{L} \to \mathbb{F}$.
2. **Combine.** Sample $r \leftarrow \mathbb{F}$ and form $u + r \cdot v$.
3. **Observe.** $u + rv$ agrees with a codeword of $\mathcal{C} = \mathsf{RS}[\mathbb{F}, \mathcal{L}, d]$ on a large set S: $|S| \geq (1 - \gamma)\ |\mathcal{L}|$.
4. **Conclude.** What is the probability that u and v do not agree with codewords on S? Define it as $\varepsilon^{\mathsf{mca}}(\mathcal{C}, 2, \gamma)$ and let coding theorists figure that out ☺.

> **Note.** Bounding this error for varying values of γ is at the core of the Proximity Prize.

Backup: mutual correlated agreement, formally

Definition (Proximity generator with mutual correlated agreement)
Let $\mathcal{C} := \mathsf{RS}[\mathbb{F}, \mathcal{L}, d]$. A sampler Gen has MCA for $\mathcal{C}$ with error $\varepsilon^{\mathsf{mca}}$ if for every $f_1, ..., f_m : \mathcal{L} \to \mathbb{F}$ and every $\gamma \in [0, 1]$:

$$\Pr_{(r_1,...,r_m) \leftarrow \mathsf{Gen}\ (m)} \left[\begin{array}{c} |S| \geq (1-\gamma) \cdot |\mathcal{L}| \\ \exists S \subseteq \mathcal{L} \text{ s.t. } \wedge\, \exists u \in \mathcal{C}, u[S] = \sum_j r_j \cdot f_j[S] \\ \wedge\, \exists j, \forall u' \in \mathcal{C}, u'[S] \neq f_j[S] \end{array} \right] \leq \varepsilon^{\mathsf{mca}}(\mathcal{C}, m, \gamma).$$

- FRI uses the Powers generator: sample $r \leftarrow \mathbb{F}$ and output $(1, r, r^2, ..., r^{m-1})$. Folding by 2 is the case $m = 2$.
- For a general folding factor m the key lemma goes through a bivariate polynomial $\hat{Q}(X, Y) = \sum_j \hat{u}_j(X) Y^{j-1}$ and $\hat{u}(X) := \hat{Q}(X^m, X)$.

Backup: round-by-round soundness

A state function $\mathsf{State}(\mathsf{x}, \mathsf{y}, \mathsf{tr}) \in \{0,1\}$ on partial transcripts; state 0 means "the prover is about to lose".

- $\mathsf{State}(\varnothing) = 1$ iff the statement is true. Prover messages cannot change the state. A complete transcript in state 0 is rejected.
- **Round-by-round soundness with errors $(\varepsilon_1, \ldots, \varepsilon_k)$**: for every partial transcript tr in state 0, $\Pr_{\alpha_i}[\mathsf{State}(\mathsf{tr} \parallel \alpha_i) = 1] \le \varepsilon_i$.

Why Fiat–Shamir needs it
A grinding prover retries each challenge: Q_i hashes at round i. It succeeds with probability at most $\sum_i Q_i \varepsilon_i$. Plain soundness assumes one challenge per round and says nothing about this prover.

Our state function
State 1 during the folding phase iff some round i had a bad fold: a large-enough $S \subseteq \mathcal{L}_{i-1}$ on which $\mathsf{Fold}(f_{i-1}, \alpha_i)$ agrees with a codeword while f_{i-1} does not. After the query phase: state 1 iff the verifier accepts.